

Introduction to Machine Vision

Lecture 1

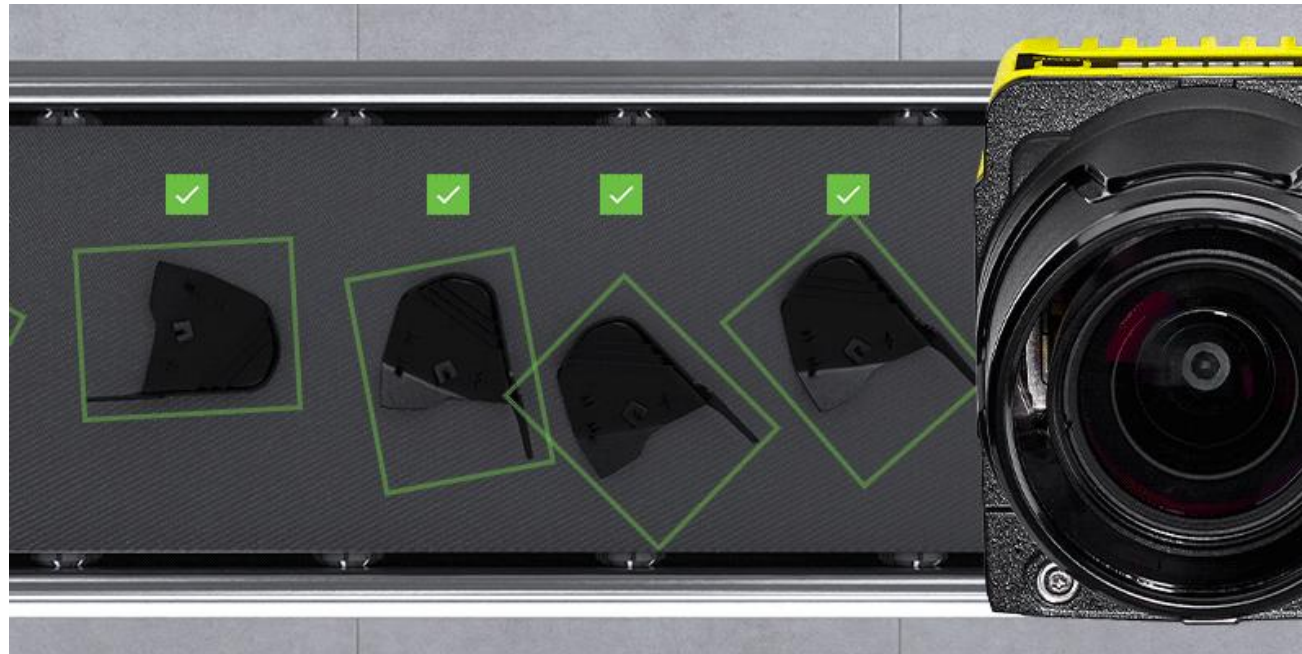
What is Machine Vision?

- Machine vision is a **technology** that enables **machines** to interpret and understand **visual information (images)** from the **physical world** to perform **tasks** or make **decisions**.
- Machine vision involves three major steps:
 1. capturing images using cameras,
 2. processing these images using algorithms, and
 3. extracting meaningful information to control a process or activity.



Human Vision vs. Machine Vision

- Unlike human vision, which is inherently **flexible** and **adaptive**, machine vision systems are designed for **specific tasks**, often in industrial settings, where **speed**, **accuracy**, and **reliability** are paramount.



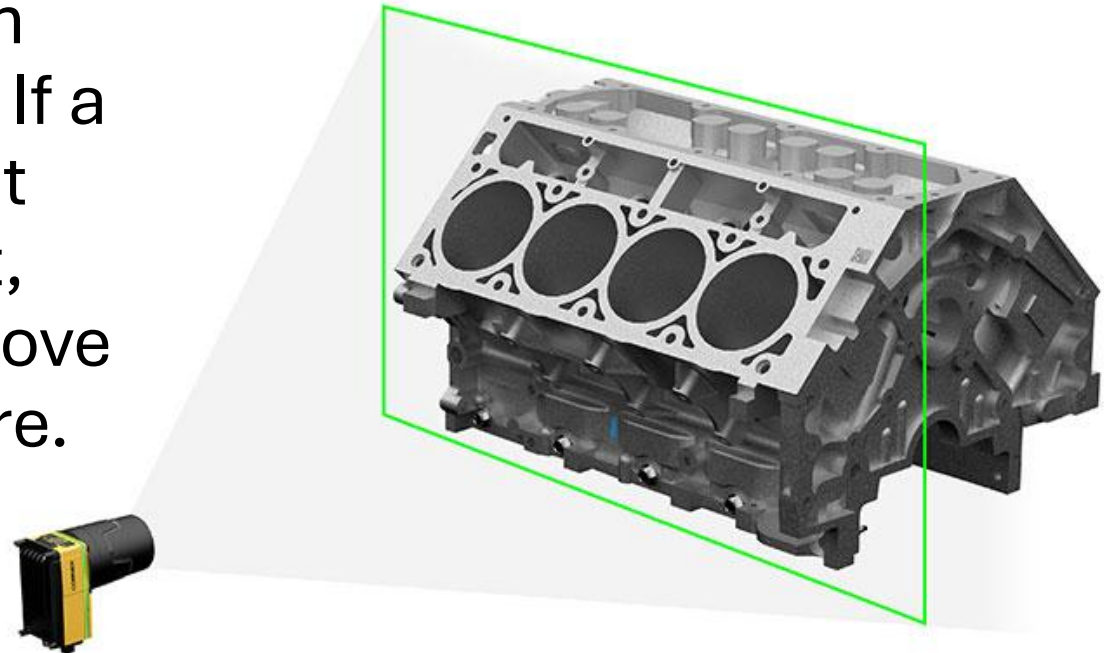
Benefits of Machine Vision

- Since it allows manufactured parts and products to be inspected, measured, and sorted, machine vision has vastly increased the power and flexibility of industrial automation.
- Sorting bell peppers with different sizes is possible without machine vision. But sorting bell peppers with different colors is impossible without machine vision.
- Machine vision does all this at high speeds and high accuracy, improving product quality and reducing waste.



Benefits of Machine Vision

- Machine vision also lets automated equipment locate objects, identify them, and save information about their material, condition, orientation, and other details for later analysis. That information is critical to factories looking for efficiency gains.
- For example, machine vision inspect an automotive part. If a fault happens more frequent than other faults on the part, engineers may need to improve the manufacturing procedure.



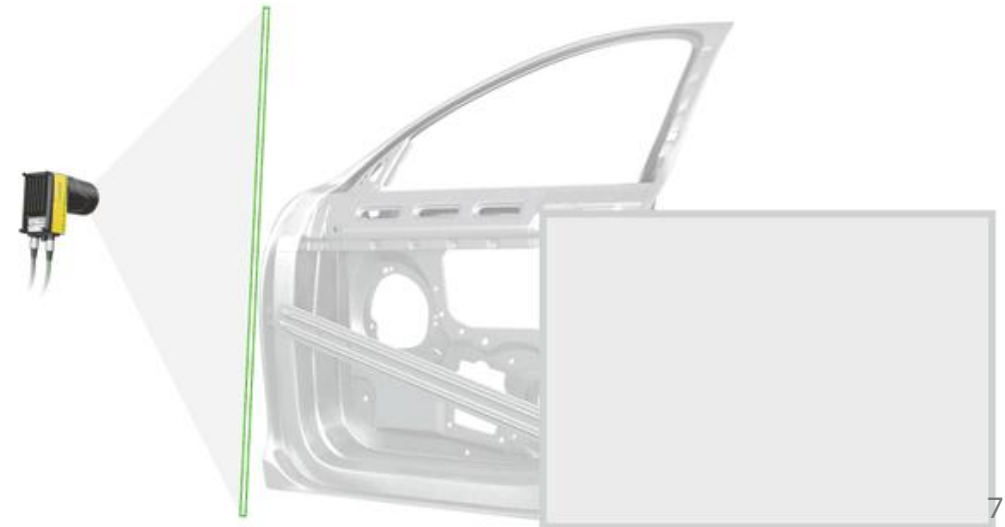
Main Types of Machine Vision Systems

- **Line-scan cameras** are used in continuous inspection applications.
- They take a wide but very thin image, typically as material moves past the scan area, and use software to reconstruct the image line by line.



Main Types of Machine Vision Systems

- **Line-scan cameras** can capture images on nonflat objects.
- They are significantly faster in these applications than standard 2D cameras because they can operate at higher frame rates.
- **Frame rate** is the number of images (frames) that a camera captures or displays per second. For example: **30 fps** → the camera captures 30 images every second.



Main Types of Machine Vision Systems

- Most machine vision uses **2D cameras**, also called **area scan systems**. These can be simple, single-purpose **sensors** or more **full-featured vision systems**.



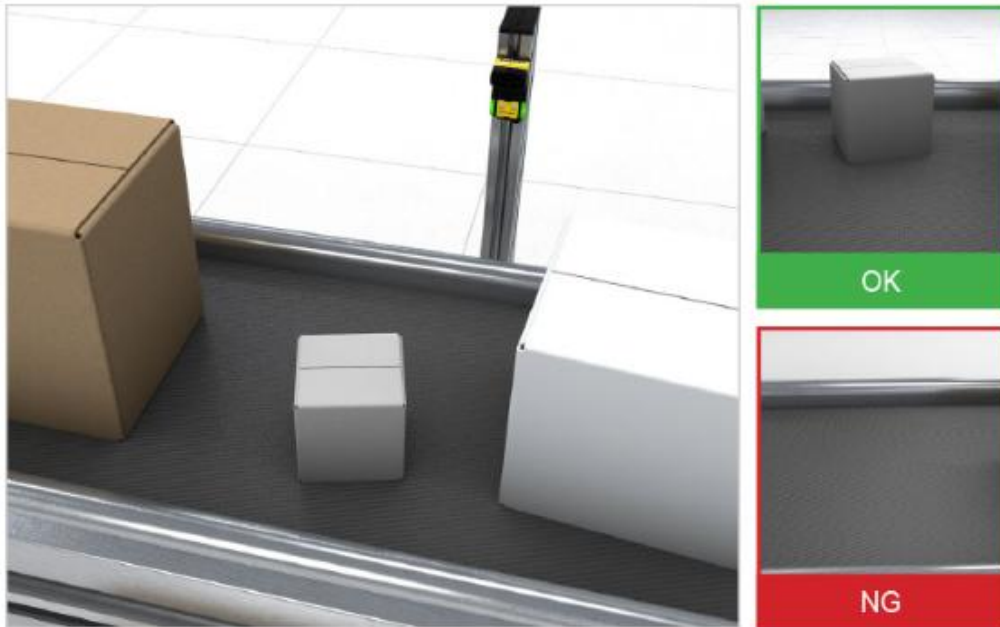
Main Types of Machine Vision Systems

- **Sensors** are lower cost, smaller, easier to deploy, and often more rugged than cameras with a wider range of features.



Main Types of Machine Vision Systems

- **Sensors** are typically used in simple machine vision applications.



Process Monitoring



Sortation

Main Types of Machine Vision Systems

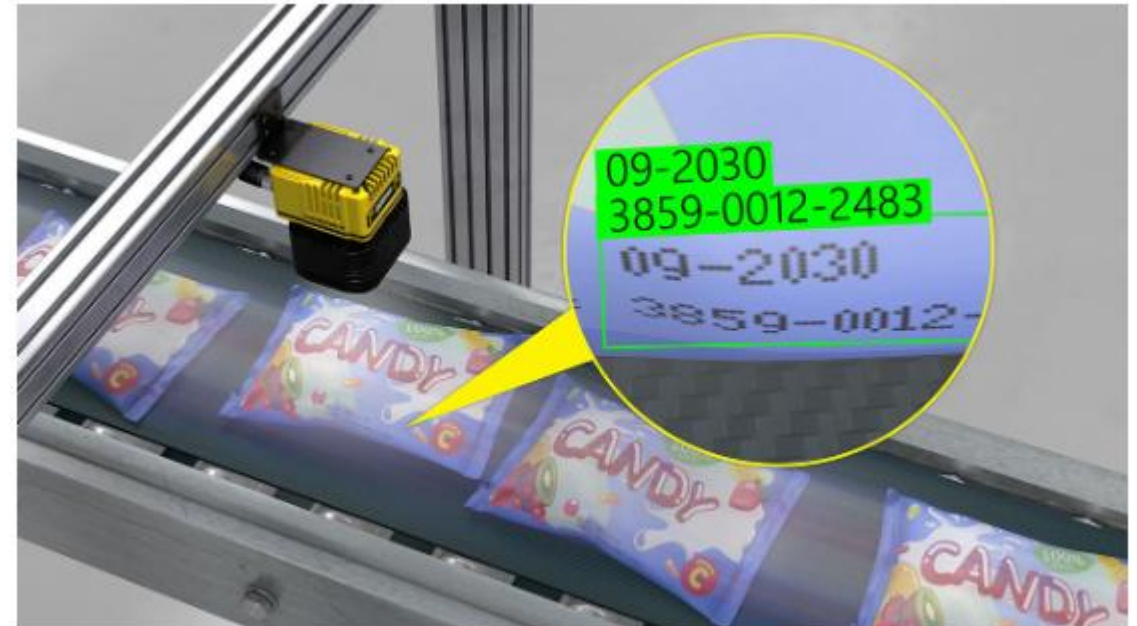
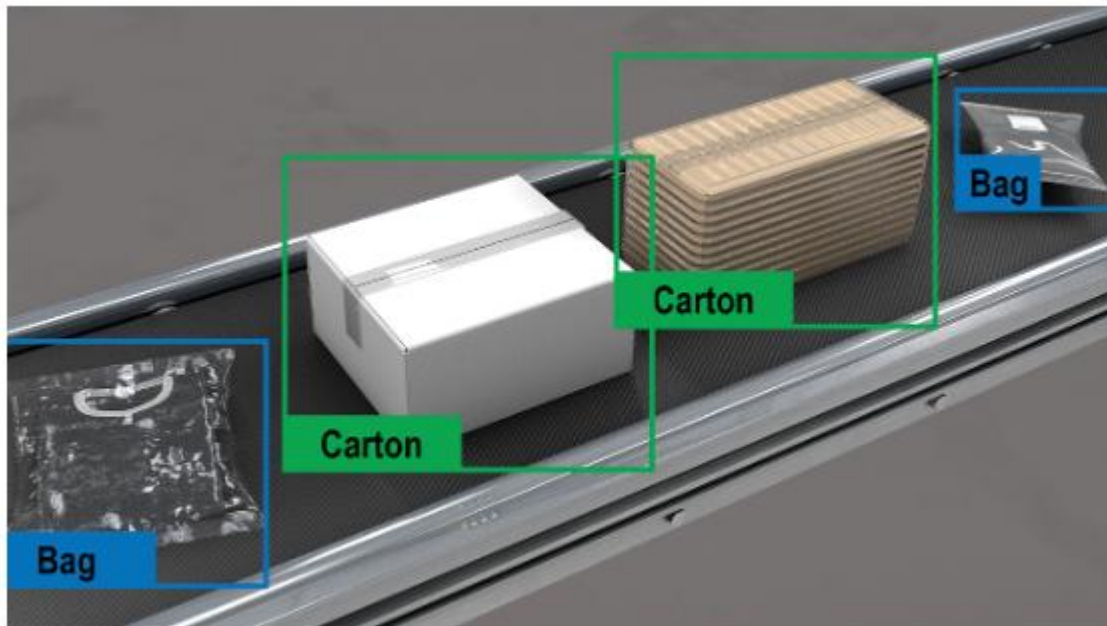
- Generally, **full-featured vision systems** are more configurable, perform more complicated tasks, and can take larger and higher resolution images.



In-Sight 7000 Series

Main Types of Machine Vision Systems

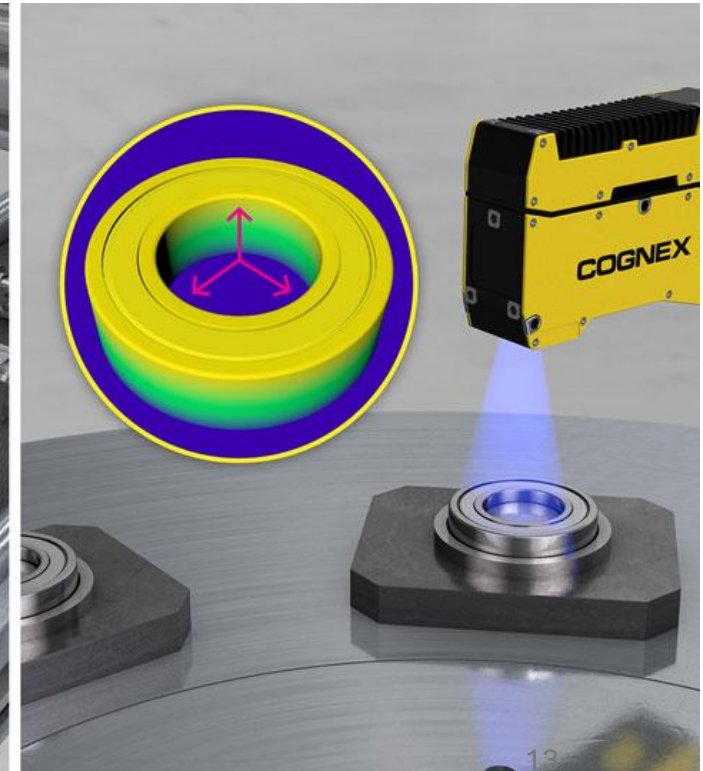
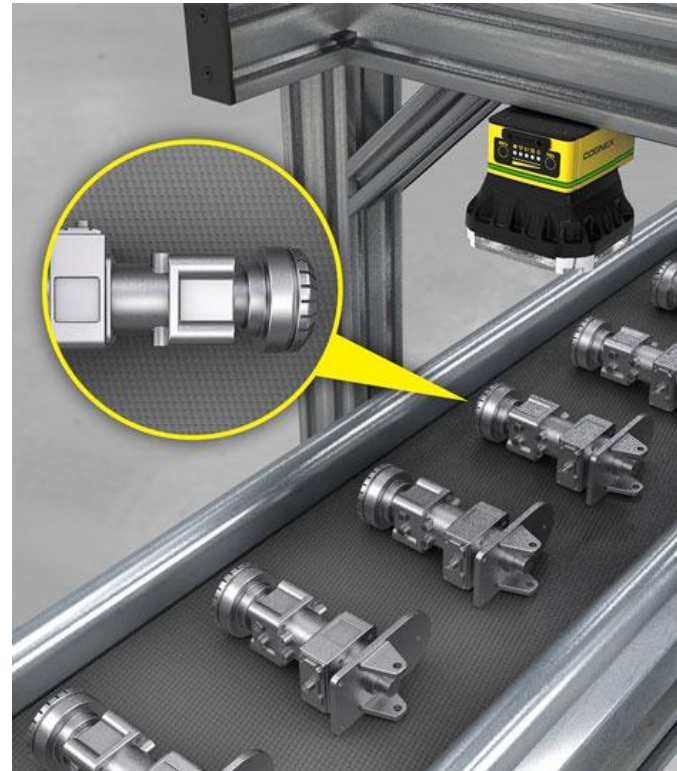
- The decision between **sensors** and **more complex machine vision systems** comes down to the task being performed, the format of the data output, cost, and ease of use.



Main Types of Machine Vision Systems

- **3D vision systems** add depth to their images, for example, by using lasers to measure distances and compute depth.
- The addition of depth can increase complexity and cost, but that additional information is critical to some applications.

2D images (left) show variations in contrast, while 3D images (right) capture a 3D map of an object

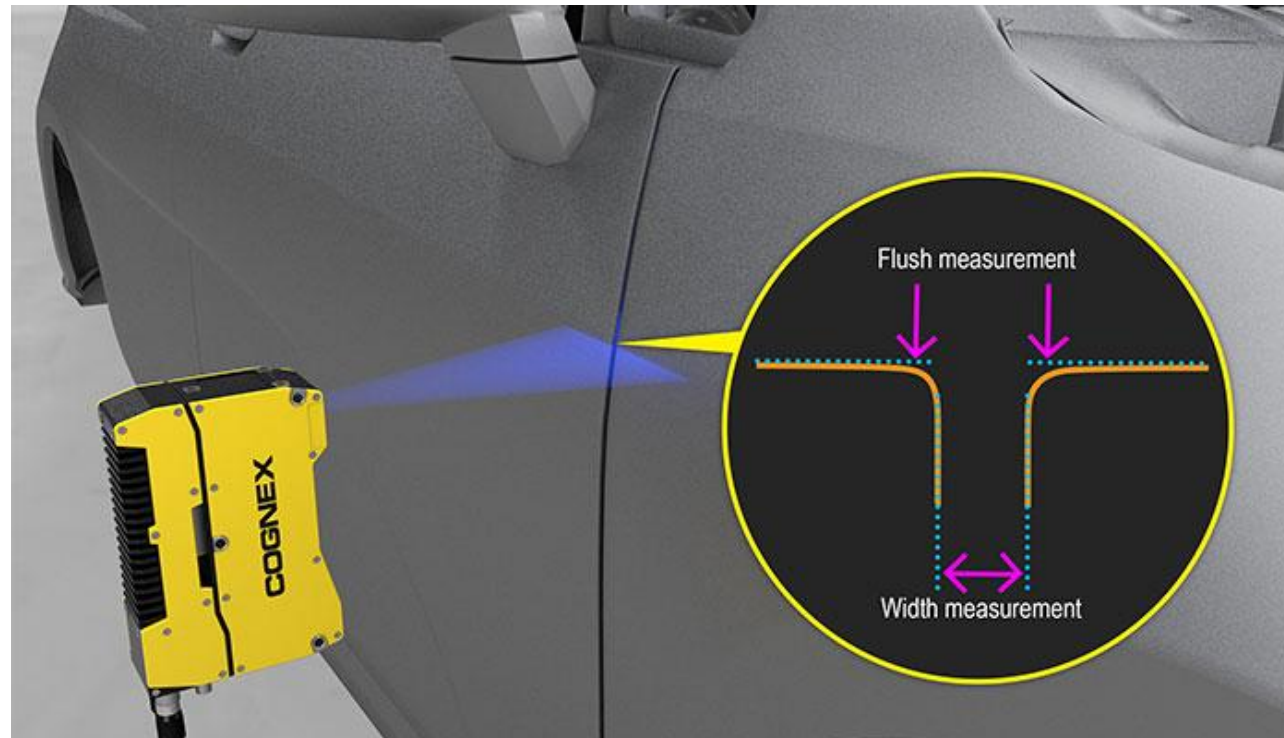


Main Types of Machine Vision Systems

- **3D vision** is essential in guiding a robot arm to precisely reach out to grab an object in the correct orientation, no matter where it is in space.
- Think of grabbing an object on a surface. If you only know its location and the orientation on the horizontal surface (top view), you'll fail to pick it up correctly because the depth from the object is unknown.

Main Types of Machine Vision Systems

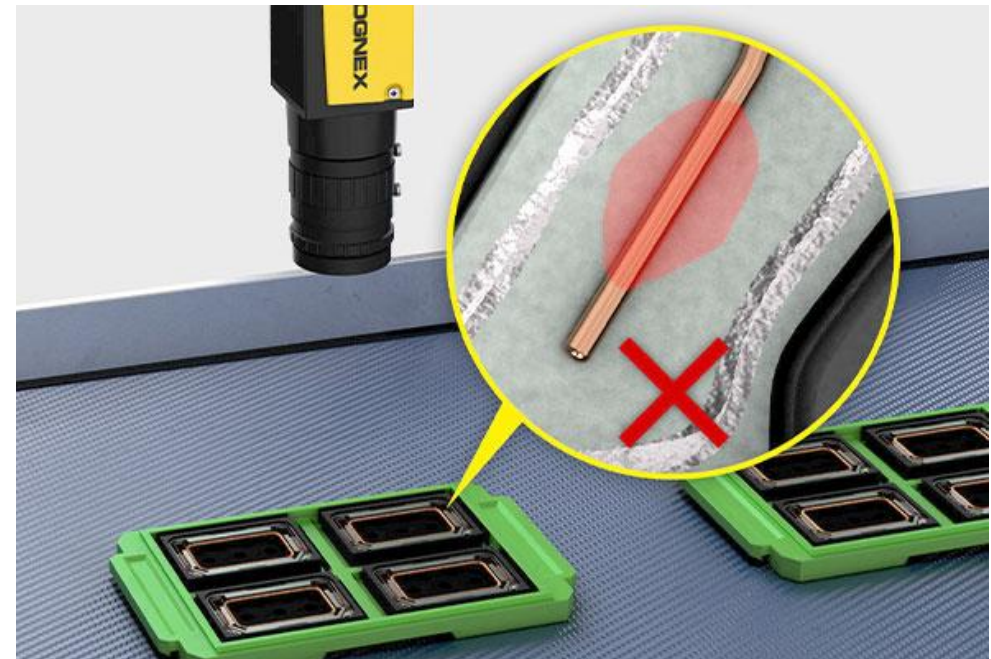
- **3D vision** is also used to gauging and measurement to check the tolerances of manufactured parts, ensuring that they fall within acceptable limits.



A flush measurement evaluates how well two adjacent surfaces align on the same level or plane

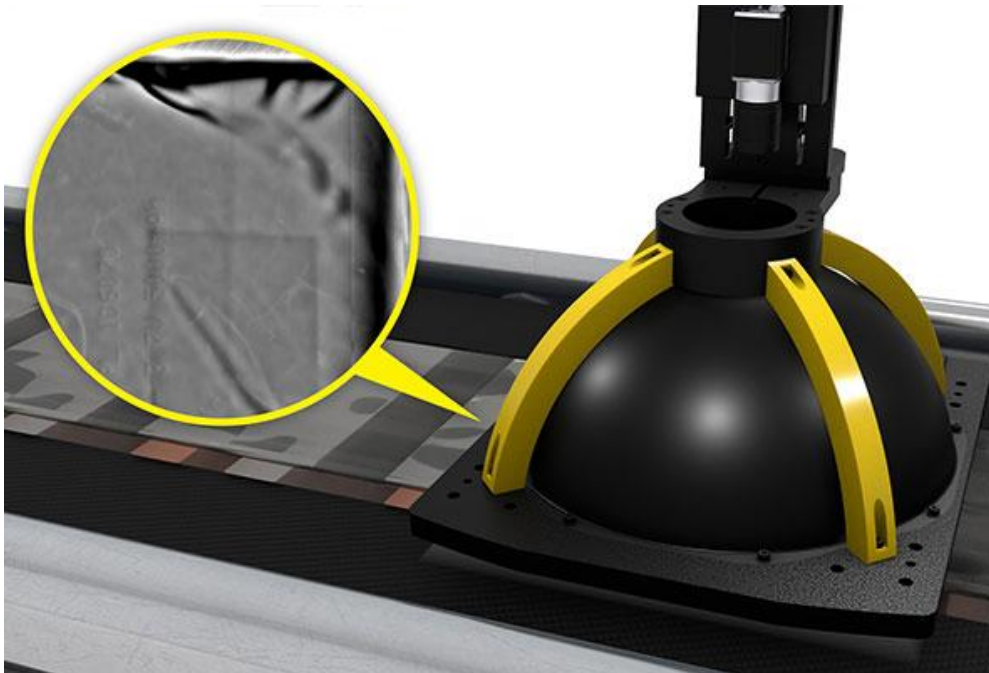
What Can Machine Vision Do?

- **Defect Detection** - Defects can occur in any part of a manufacturing process, from problems with the quality of raw materials or parts through final inspection. Inspections to catch defects have traditionally been done manually, by trained workers.
- Examples 1: Looking for loose connectors, poorly soldered wires, bad seams, or improperly crimped tubes on a printed circuit board.



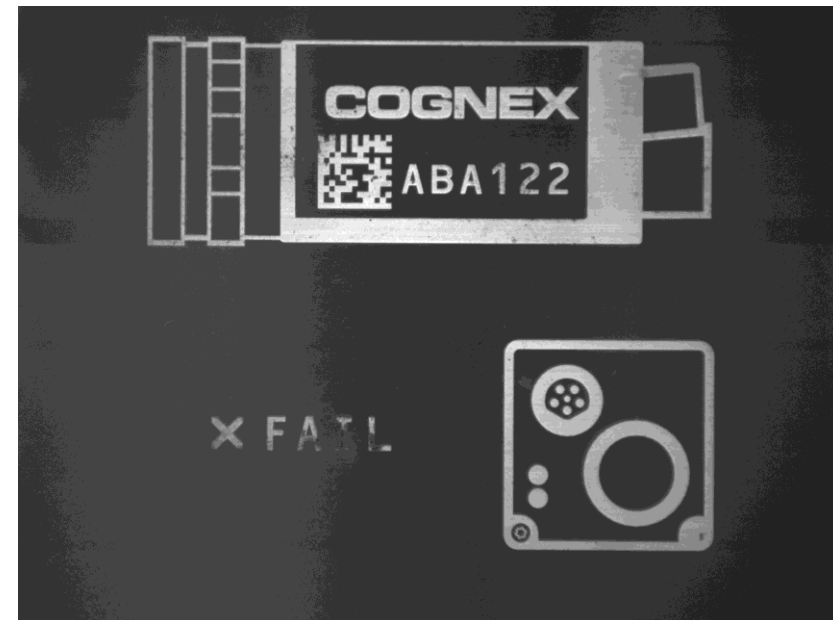
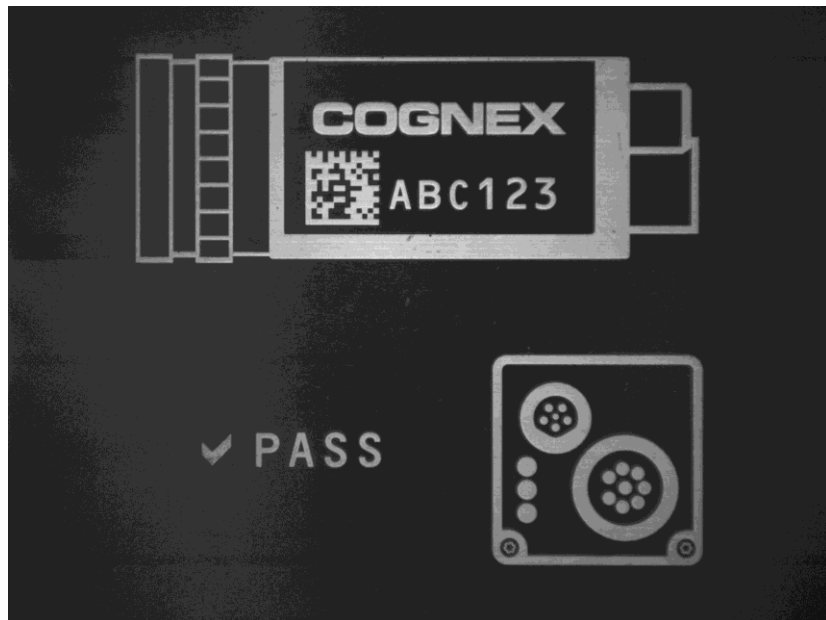
What Can Machine Vision Do?

- Example 2: Detecting flawed photovoltaic cells in solar panels or defects in semiconductor wafers or EV battery assemblies.
- Example 3: Finding contaminants or other problems in food products.



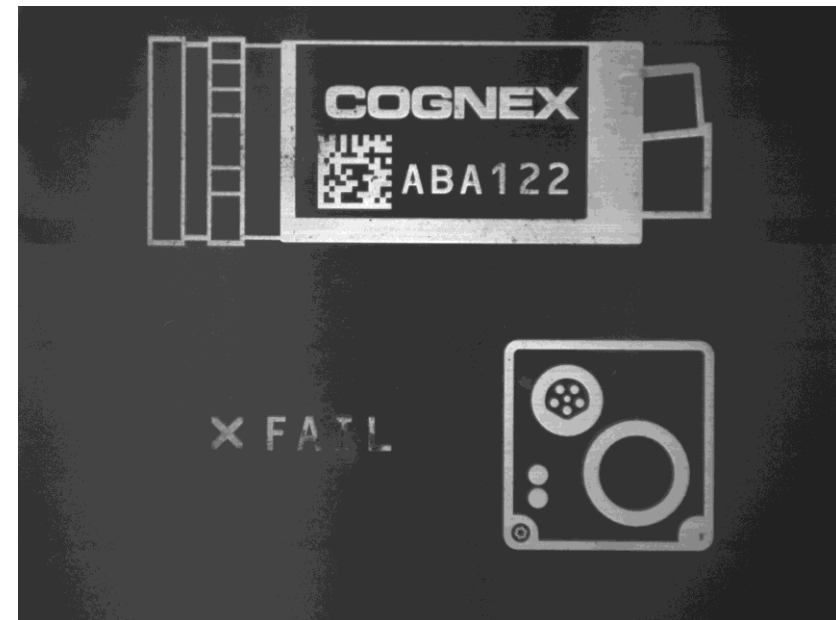
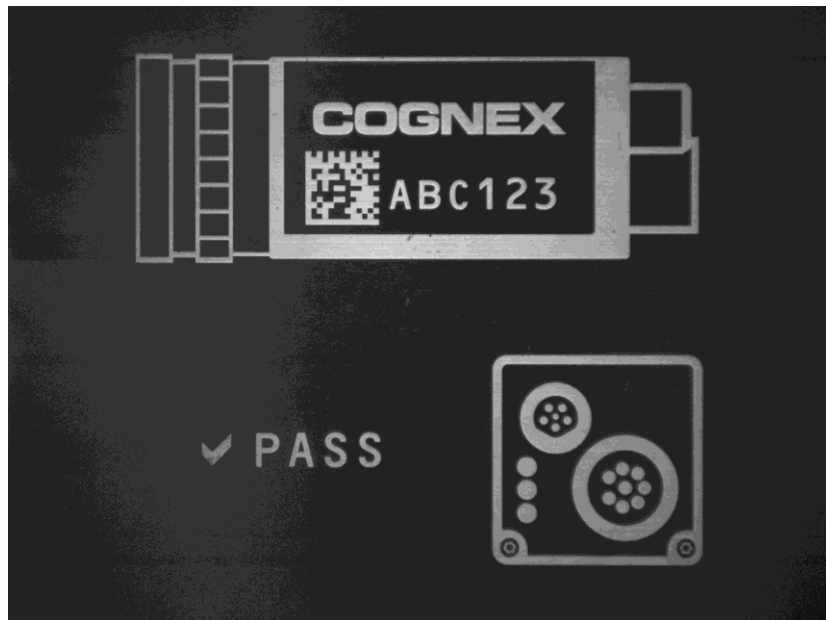
What Can Machine Vision Do?

- **Object detection and counting** - Determining the presence or absence and counting objects is a widely used function in inventory management, on production lines, and before releasing or accepting shipments.



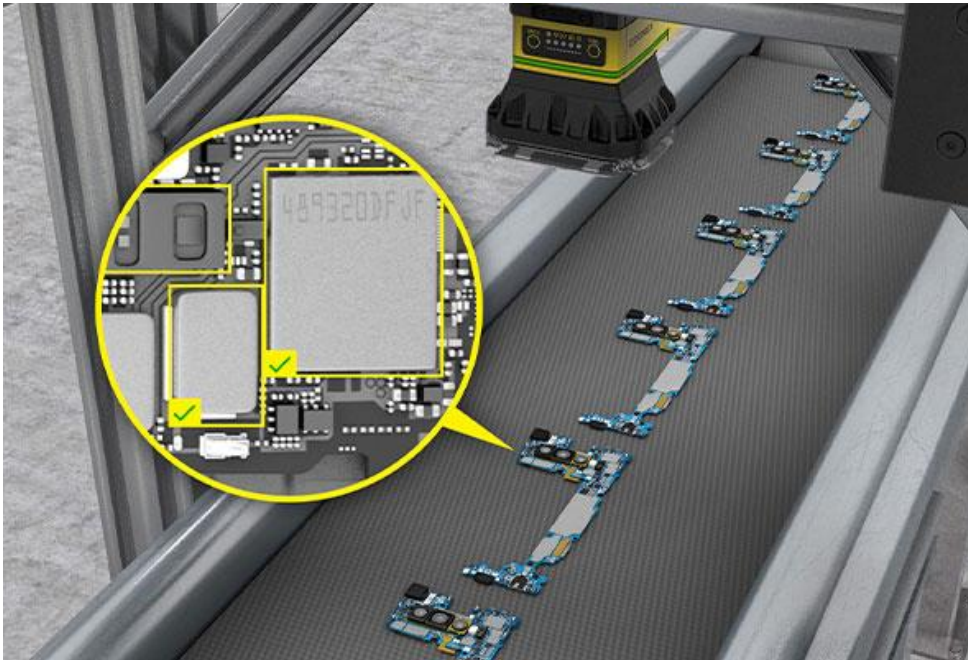
What Can Machine Vision Do?

- Both manual inspection and mechanical counting are slow and prone to error, compared to high-speed and consistently accurate machine vision systems.



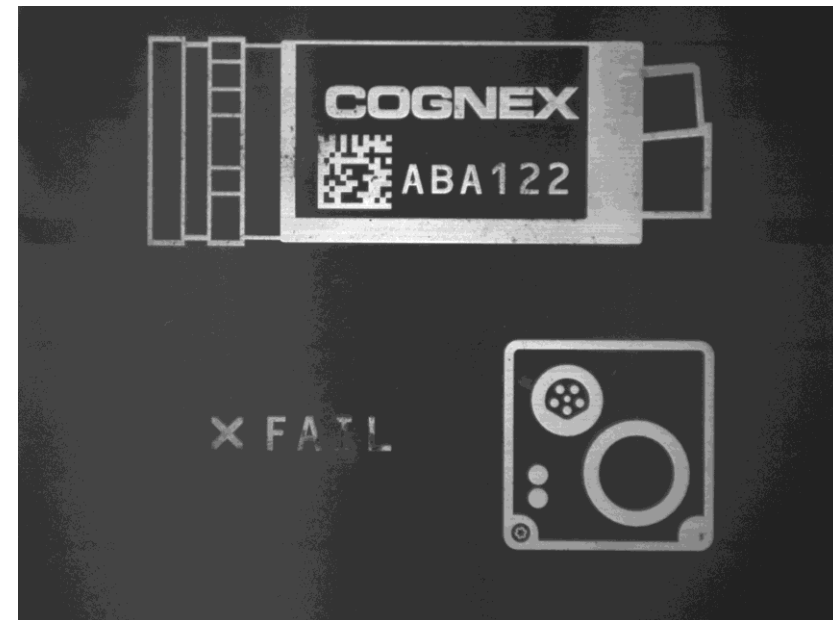
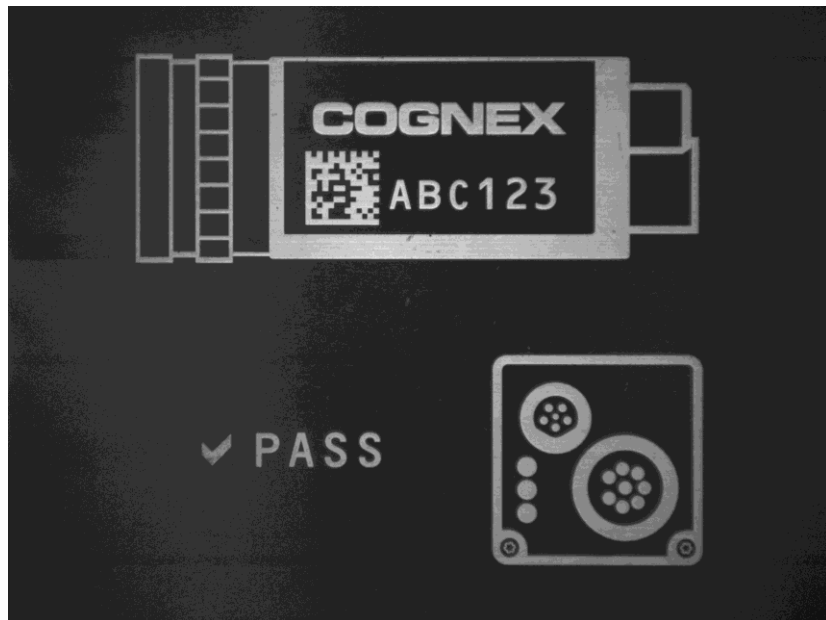
What Can Machine Vision Do?

- Example 1: Confirming the presence of all electronic components on a printed circuit board.
- Example 2: Verifying the presence of components such as clips, screws, springs, labels, seals, manuals, inserts, or accessories.



What Can Machine Vision Do?

- **Measuring/gauging** - Precise manufacturing requires accurate measurement of distances, areas, diameters, and more. Manual measurement with gauges, calipers, or inspection jigs is slow and introduces inconsistencies.



What Can Machine Vision Do?

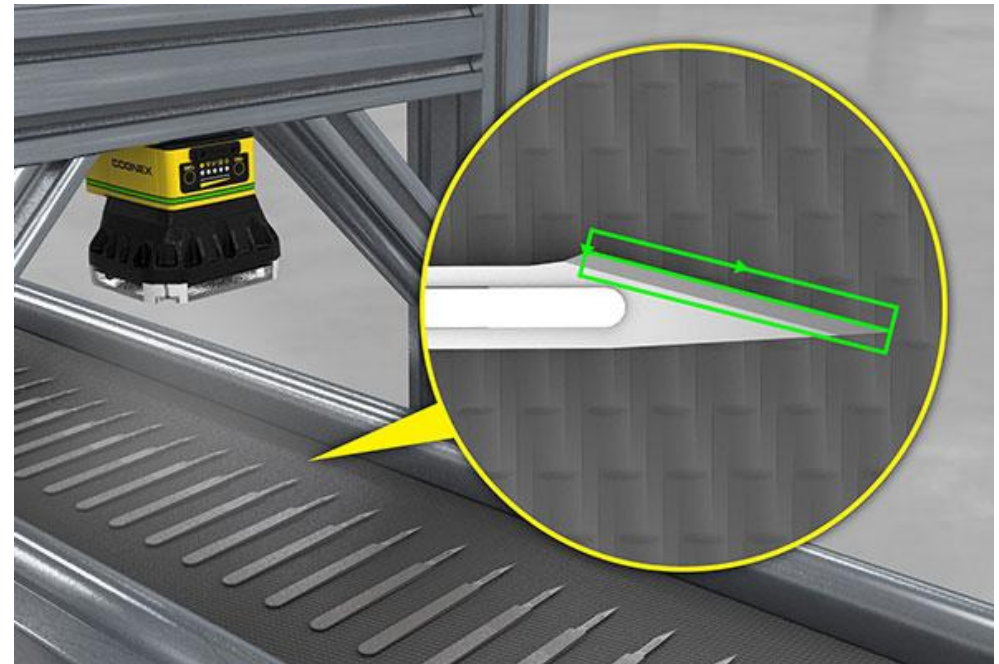
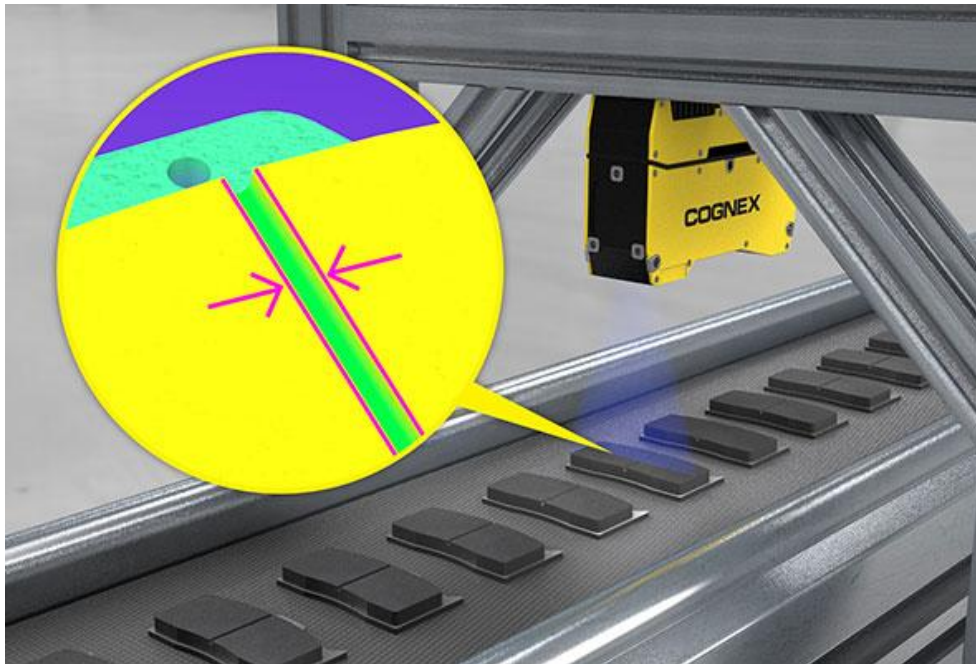
- Machine vision consistently and accurately measures down to the micron level, while parts are on the line, at high speeds. Each image and its associated data can be stored in case of warranty or compliance issues.

Machine Vision spot even the smallest anomalies in Amazon delivery vans—from tire deformities and undercarriage wear to bent or warped body pieces



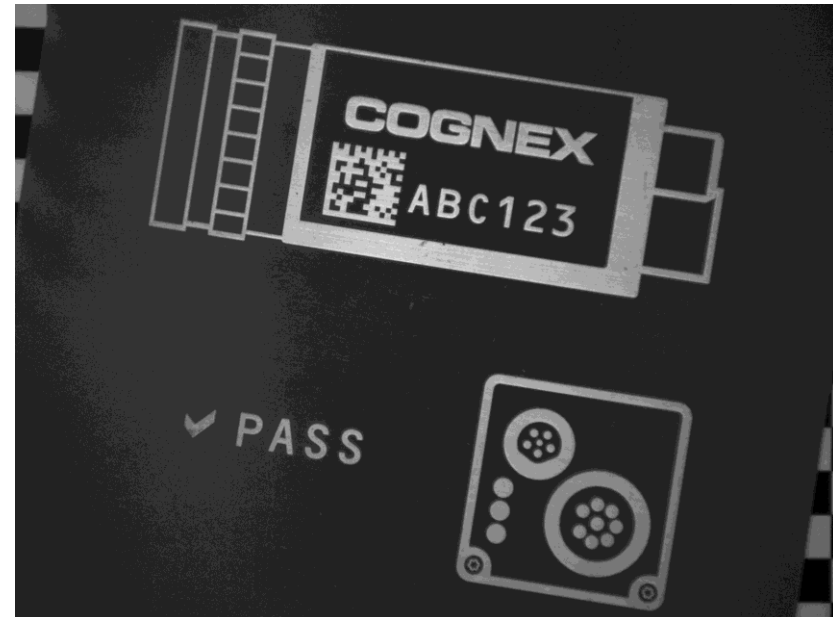
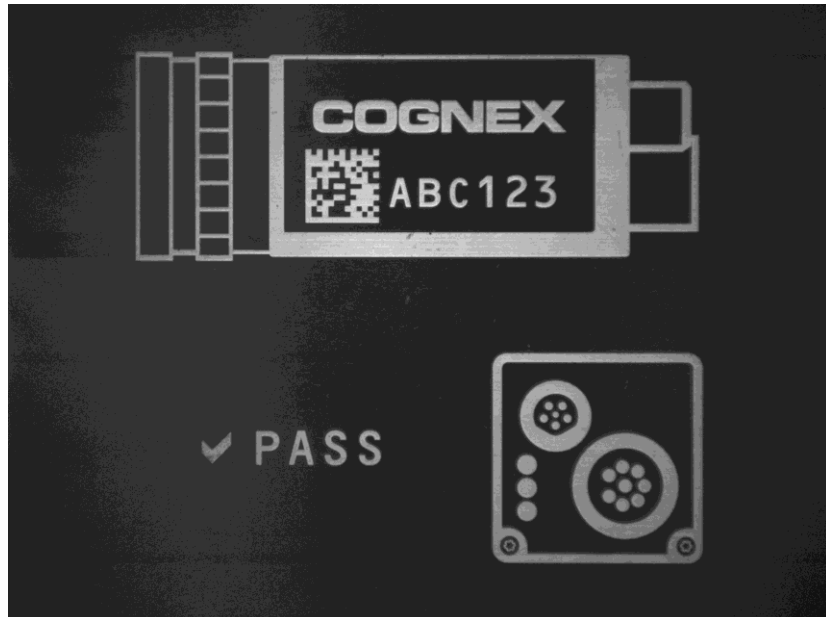
What Can Machine Vision Do?

- Example 1: Capturing the dimensions of cast and injection molded parts.
- Example 2: Measuring the roundness and angle of tips on parts .



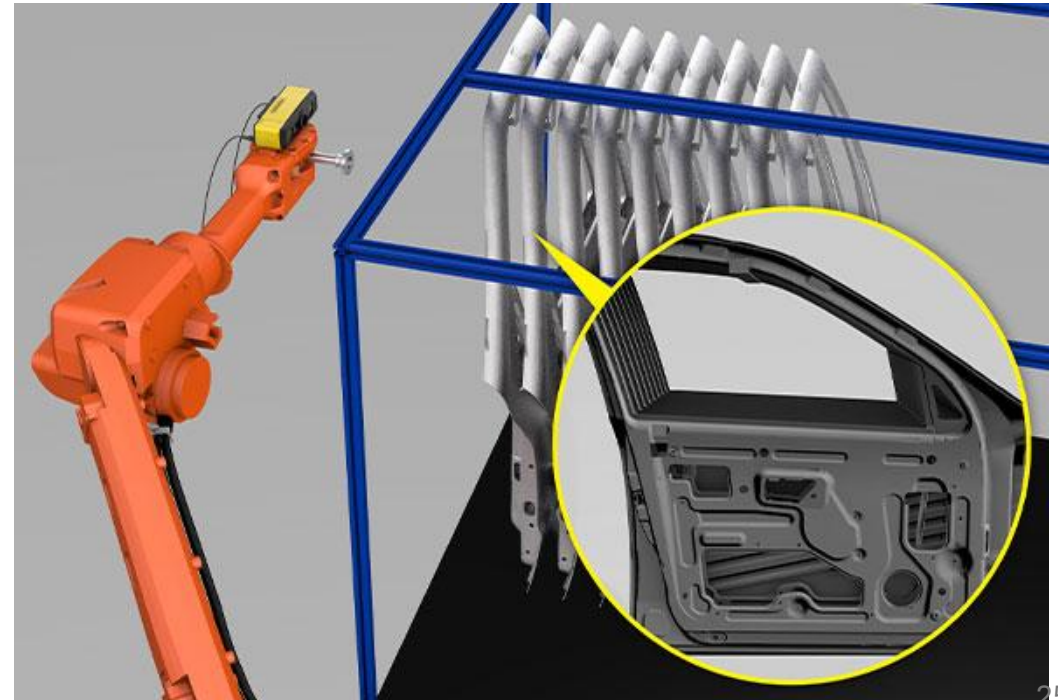
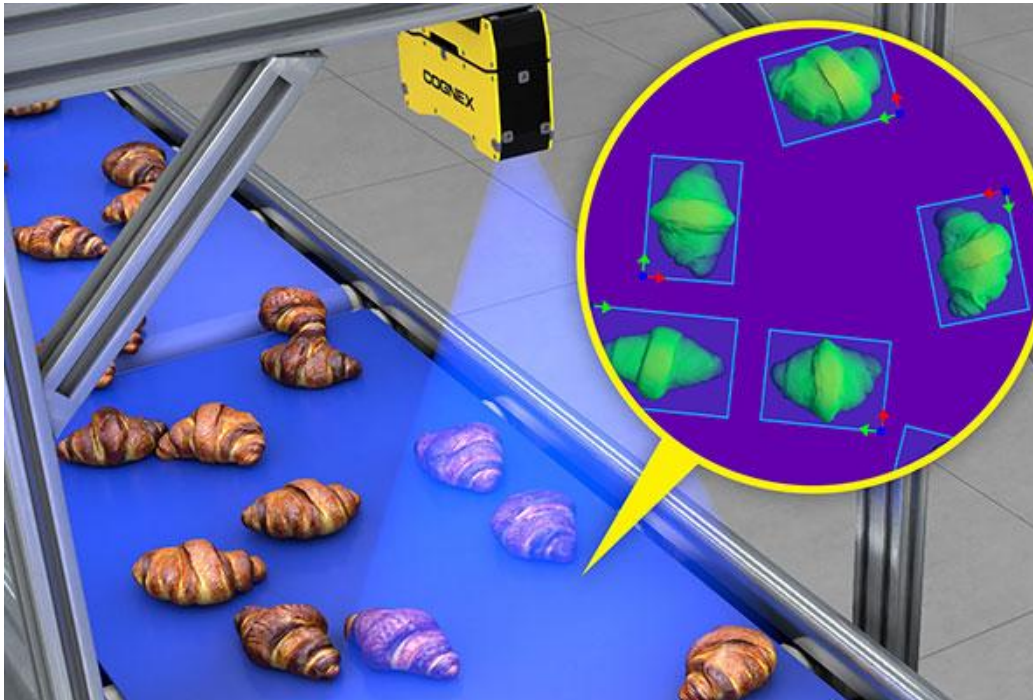
What Can Machine Vision Do?

- **Locating/guiding/positioning** - Functions such as assembly, pick-and-place, and inspection depend on machine vision's ability to locate a part, whether on a conveyor or in a bin. Machine vision guides robots in picking and placing parts and ensures accurate micron-level positioning in precision assembly.



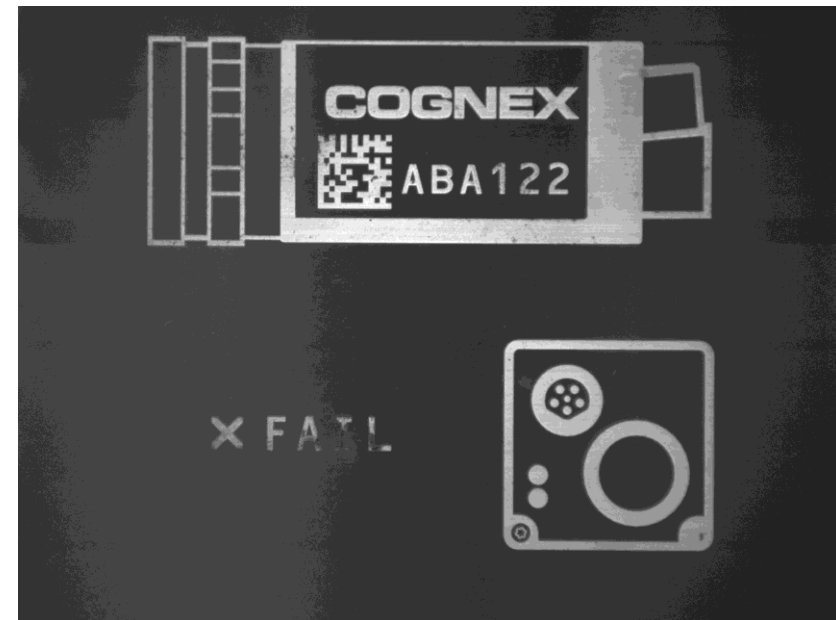
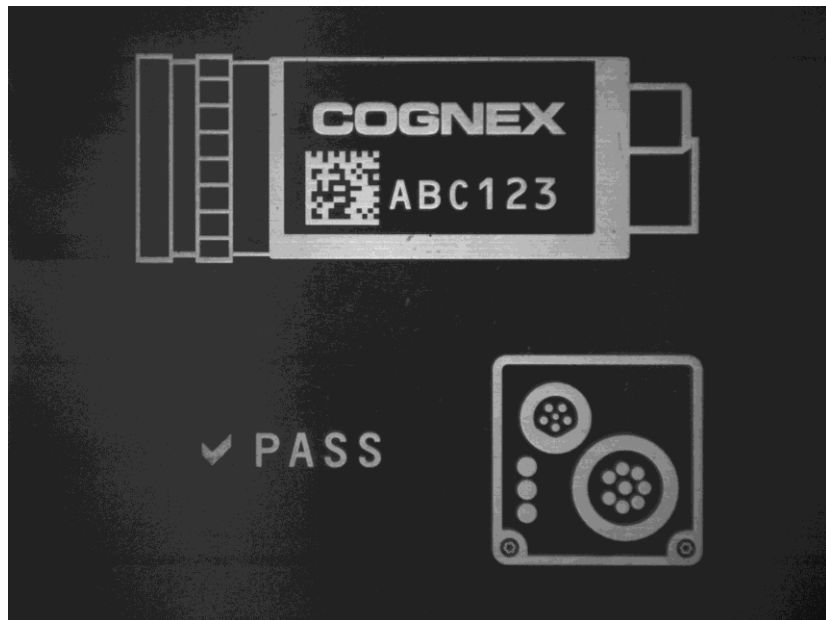
What Can Machine Vision Do?

- Example 1: Locating parts on a conveyor for inspection.
- Example 2: Guiding robots on automated automotive assembly lines.



What Can Machine Vision Do?

- **Barcode reading** - Barcodes are used to track and identify raw materials and finished goods throughout the supply chain. They include 1D and 2D codes, such as universal product codes (UPC) on retail products and data matrix codes on packages.

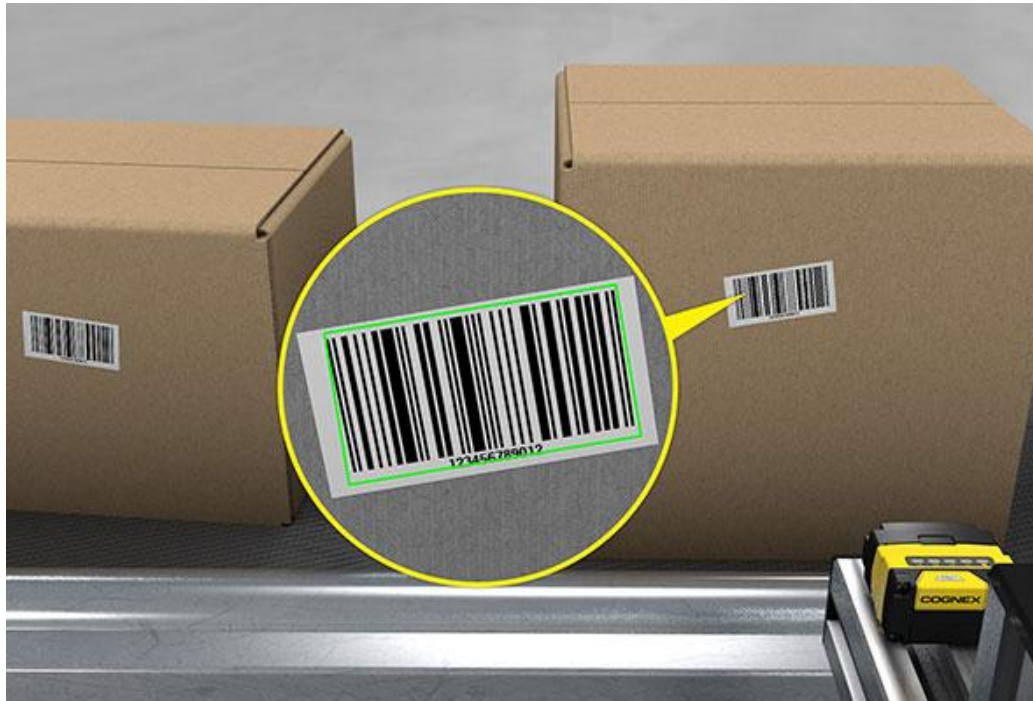


What Can Machine Vision Do?

- Barcode readers process information at high speeds, and correctly read codes that are partially torn, obscured, smeared, or distorted.
- Multiple camera systems can scan from multiple sides of a part or package at once to increase read rates when not all codes are oriented the same way.

What Can Machine Vision Do?

- Example 1: Tracking packages as they travel through a logistics warehouse.
- Example 2: Providing accurate traceability of medical supplies.

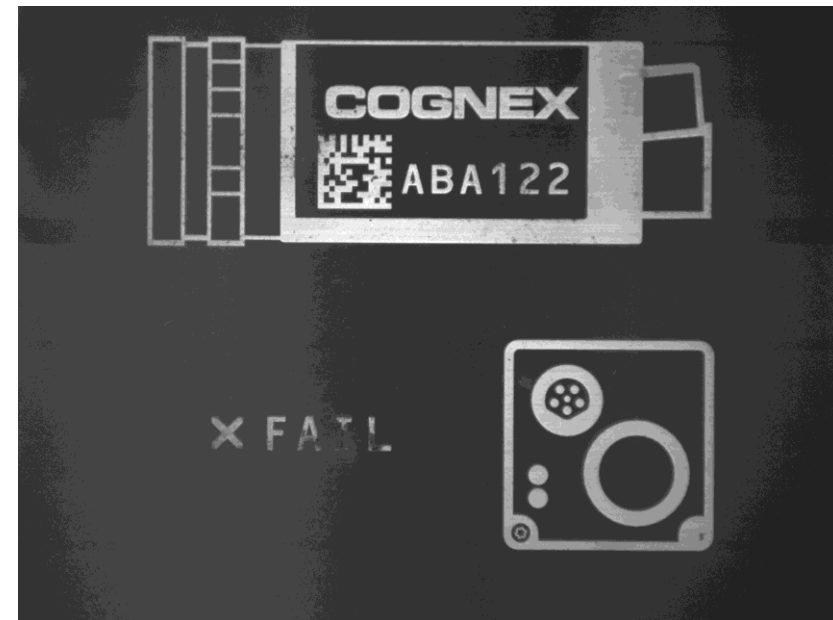
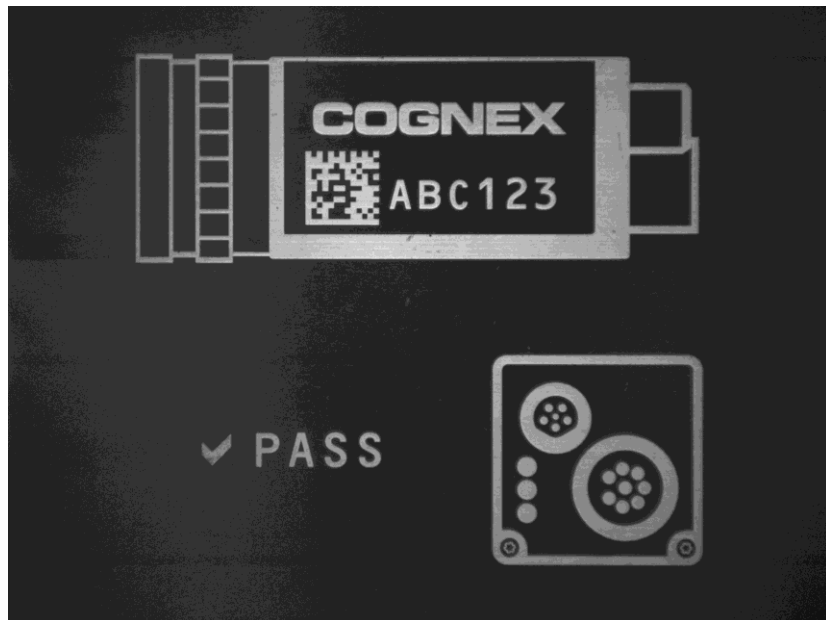


What Can Machine Vision Do?

- **Reading text: OCR/OCV** - Barcodes are everywhere in modern industry, but human-readable printed text is still essential, particularly for retail, manufacturing, pharmaceutical, and food and beverage supply chains, to identify **best-before dates, lot numbers**, and other important information.

What Can Machine Vision Do?

- **Optical character recognition (OCR)** and **optical character verification (OCV)** both identify and interpret text in images, with one key difference. OCR reads text to trigger another action, while OCV is used to verify the quality of text against a known standard.



What Can Machine Vision Do?

- Example 1: Classifying a part based on the characters printed on it (OCR).
- Example 2: Checking if a best-before date or lot number is printed correctly (OCV).



How Do Machine Vision Make Decisions?

- **Rule-based systems** follow user-programmed, step-by-step instructions to interpret images and make decisions.
- In contrast, artificial intelligence or **AI-powered systems** use a database of reference images to “learn” how to make decisions.

How Do Machine Vision Make Decisions?

- **Rule-based machine vision** is still the prevalent technology, however, various types of AI-powered machine learning have become capable and flexible enough to take over in some applications.
- Often, a combination of rule-based and AI-powered machine learning can provide the most efficient solution.

How Do Machine Vision Make Decisions?

- **Rule-based machine vision** uses specific **if-then** rules to make decisions about an image.
- The rules are used to create programs that tackle specific jobs, such as detecting the edge between light and dark areas or measuring the distance between two points.

How Do Machine Vision Make Decisions?

- “If the bottle is filled to at least this level, let it pass.” The rules are programmed by vision engineers with deep knowledge of the best combination of techniques to get the desired output.
- These rules can then be used one after another on an image, in what is called “tool chaining,” to perform a wide range of sophisticated tasks.
- They’re also very good at high-speed, high-accuracy inspections on products or parts that are predictable and consistent.

How Do Machine Vision Make Decisions?

- Example:

Find a line							Edge_1
Image	Row	Col	Angle				Calib
Fixture	0.000	0.000	0.000				
	Row	Col	High	Wide	Angle	Curve	
Region	401.327	42.743	60.782	44.794	36.716	0.000	
Tool Enable	Include	In J	Edge Contr	Edge Trans	Find By	Angle Range	
1.000	☒	25	Both	Best Sc	10		
Result	Invert	Show Mid-Point		EnabledSt			
Present	☐	☒		1.000			
Focus	Tool Pass	Tool Fail	Status	Pass/Fail	ShowGrapl	ShowResu	
0.000	1	0	1	1.000	1.000	1.000	
	Row0	Col0	Row1	Col1	Point	Plot	
Edges	387.007	61.944	435.401	97.416	411.204	79.680	
	8.785	59.087	14.126	66.581	11.454	62.831	
Plot	Point	Point	Point				
	Passes	Failures	Errors	Total			
Count	1	0	0	1	0.000		
					0.071	End	



How Do Machine Vision Make Decisions?

- **Deep Learning** uses AI to power machine vision systems by using examples to train the software with labeled images until it can make distinctions on its own.
- This training mimics how humans learn, helping the system learn how to make accurate decisions without being thrown off by irrelevant variation.
- Operators train deep learning systems with hundreds of labeled images (or more), such as a variety of possible defects in a manufactured part, or several different assemblies that need to be classified.

How Do Machine Vision Make Decisions?

- One significant improvement over rule-based systems is that deep learning systems learn to distinguish between a wide range of real defects and merely cosmetic variations without needing to be shown every possible result – something time-consuming or even impossible to accomplish using rule-based programming.

How Do Machine Vision Make Decisions?

- In contrast to rule-based systems, deep learning systems don't require experienced machine vision programmers or expertise to set up. They do need someone with good knowledge of the products being inspected to provide hundreds of images of both defective and good components.

How Do Machine Vision Make Decisions?

- Deep learning excels at assembly verification, defect detection, and classification of complex parts that vary unpredictably. It's also good at providing accurate results even when parts are reflective or images are distorted.



How Do Machine Vision Make Decisions?

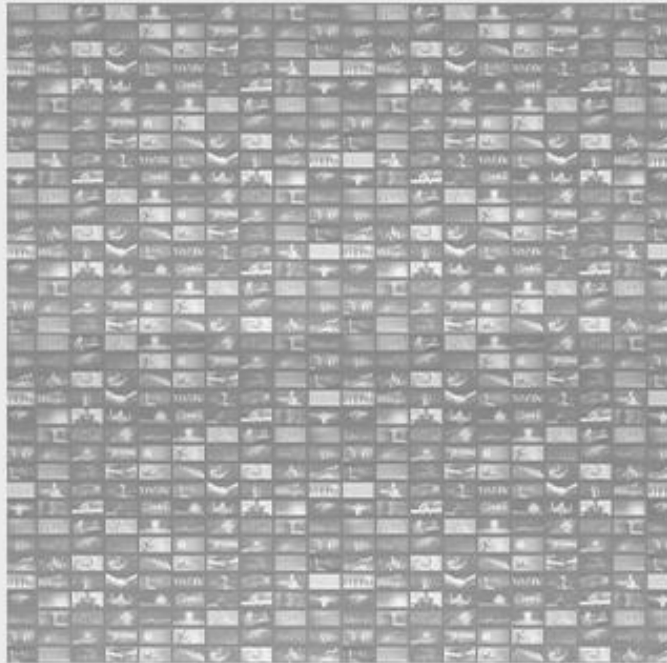
- **Edge Learning** is another type of AI, optimized to meet the needs of industrial automation. It comes **pre-trained** to solve the types of problems typically faced by industrial automation.
- A pre-trained AI model is a machine learning system that has already learned basic language, patterns, and general knowledge from a massive dataset before you use it.
- As a result of the pre-training, an edge learning system can be trained on a specific industrial inspection problem with **as few as five images**, making it even easier and faster to deploy than a deep learning system.

How Do Machine Vision Make Decisions?

- Edge learning enables line engineers to quickly implement an optimized classification or defect detection function on their line, without specialized training.
- Since it doesn't require a sophisticated processor, it can be quickly deployed where it is needed.

How Do Machine Vision Make Decisions?

- The advantages of edge learning lie in its speed, low demand on computing resources, and flexibility, as well as its user-friendliness.

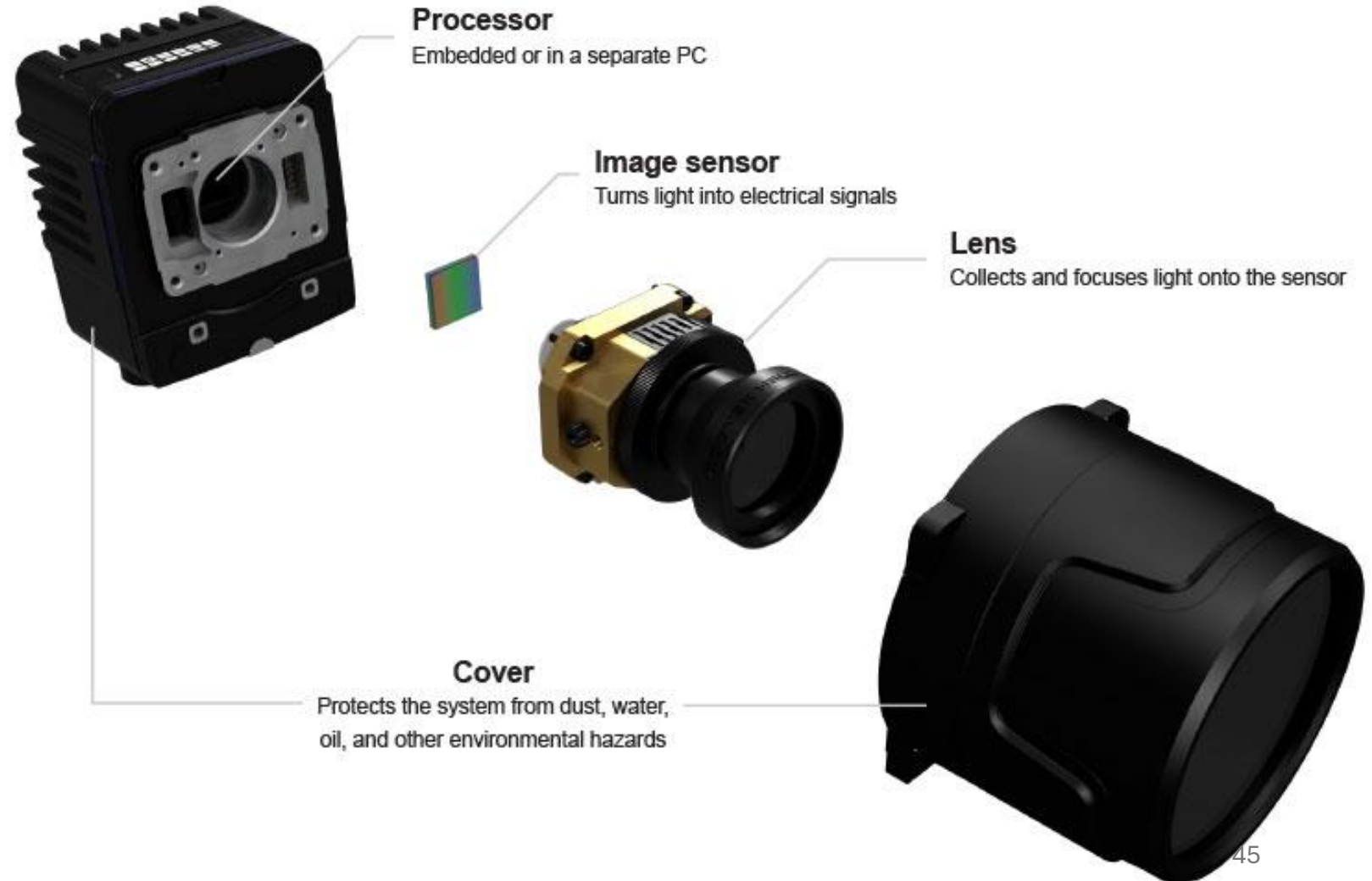


Comparing Rule-Based and AI-Powered Systems

- The decision between rule-based, deep learning, and edge learning systems mainly comes down to the type of application you're implementing.
- **Rule-based vision tools** are effective at a variety of specialized tasks with very consistent targets, such as location, measurement, and orientation.
- **Edge learning** excels at repeatable tasks with moderately consistent targets.
- **Deep learning** shines when looking for complex defects or analyzing images with significant variation.

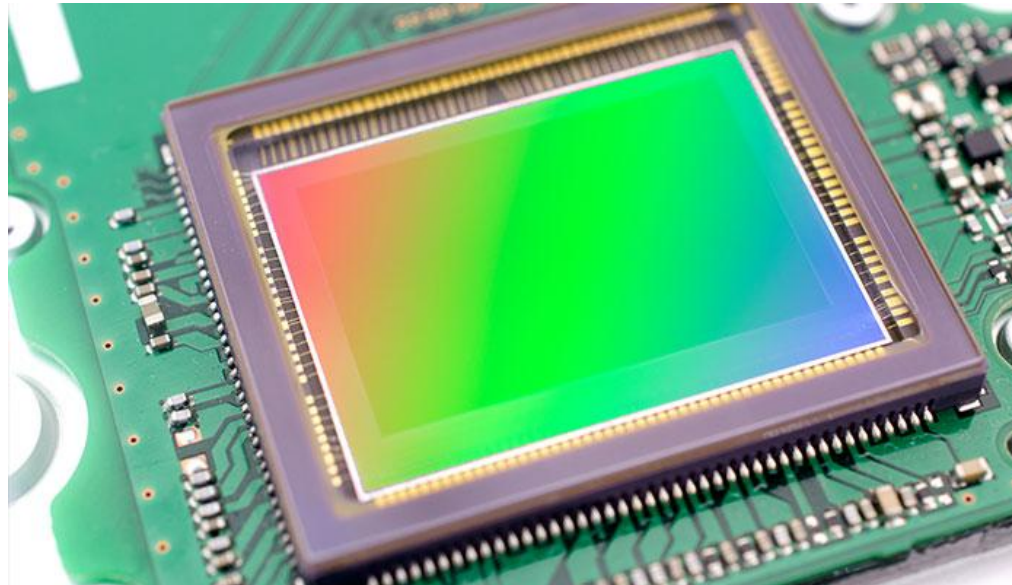
Machine Vision Hardware

- The hardware components of a vision system include the image sensor, lens, cover, and processor.



Machine Vision Hardware

- At the center of any vision system, an image sensor converts light energy into electrical signals that can be analyzed by software.
- Image sensors are solid-state semiconductor chips comprising millions of photodetectors called **pixels**, which convert light into electrical signals.

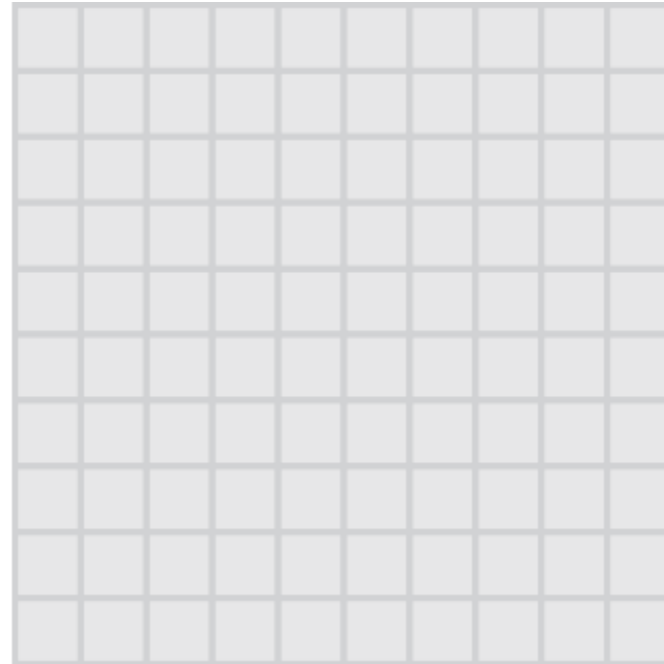


Machine Vision Hardware

- The two main technologies used for image sensors are charge-coupled device (CCD) or complementary metal oxide semiconductor (CMOS).
- **CMOS technology** is newer and taking an increasing share of the market, thanks to lower costs and faster processing speeds, but **CCD sensors** are still used in applications that require extremely high-quality images.

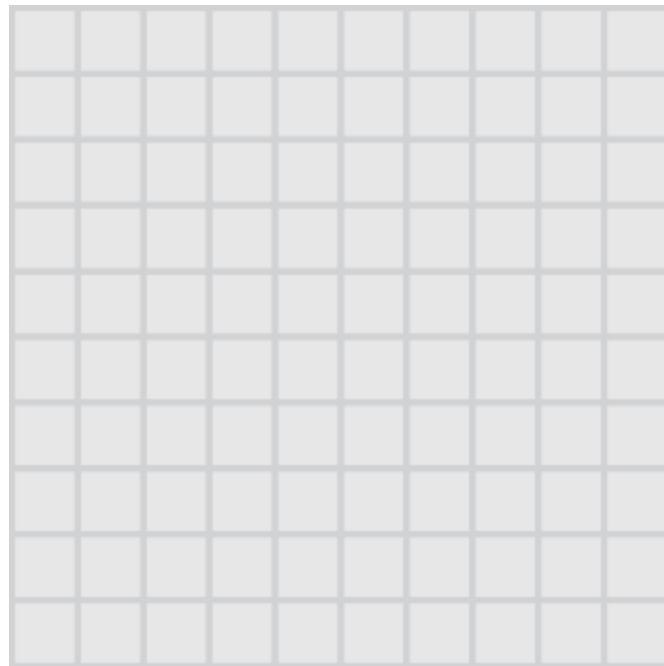
Rolling and Global Shutters

- Machine vision cameras use electronic shutters to determine how the pixels are exposed to light.



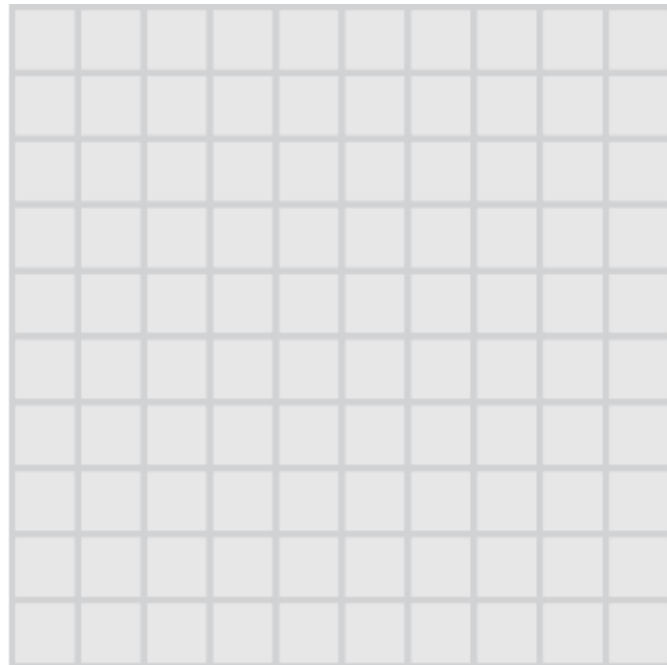
Rolling and Global Shutters

- A **rolling shutter image sensor** exposes lines of pixels sequentially. This kind of sensor is lower cost, but if the target being imaged is moving rapidly, the image can be distorted.



Rolling and Global Shutters

- A **global shutter image sensor** can expose all its pixels at once. They can be more complex to manufacture, but prices have come down and they are widely used in industrial applications.



Machine Vision Lenses

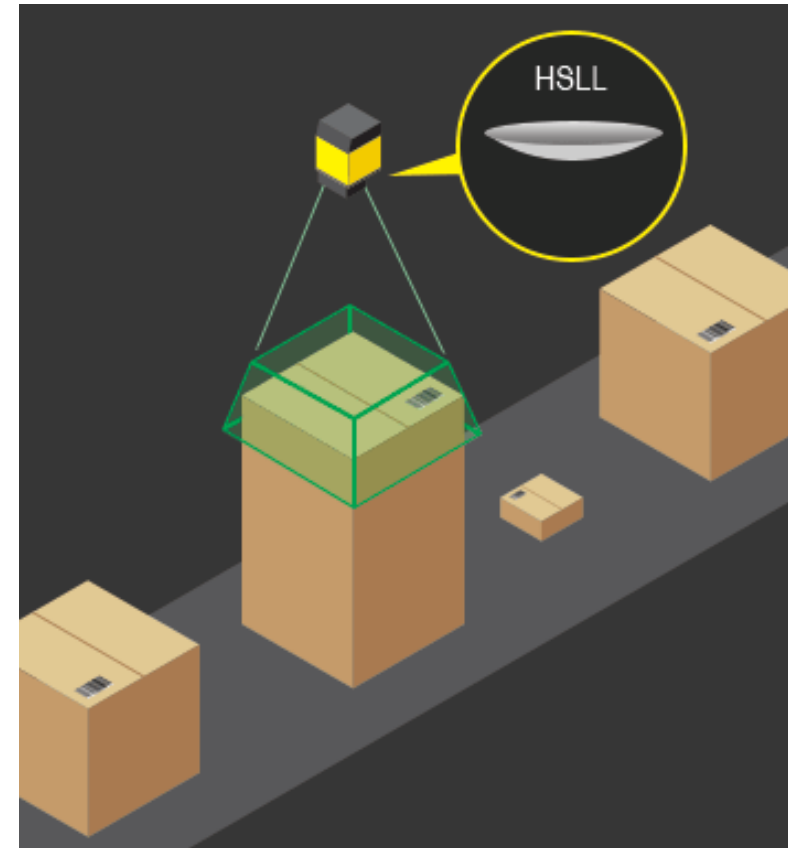
- Lenses collect and focus light onto image sensors. They are matched with the sensor so they completely illuminate the photosensitive area with a resolution equal to or greater than the sensor's pixel size.
- **Autofocus lenses** adjust their focus automatically as needed, saving time if frequent changes are expected.

Machine Vision Lenses

- **Manual lenses** need to be refocused if the objects being observed change, or if the camera moves.
- But this can be time-consuming if done frequently, or if there are many cameras that need to be adjusted. In these situations, autofocus lenses are optimal.

Machine Vision Lenses

- **Liquid lenses** are a specialized subset of autofocus lenses that adjust their shape on the fly to maintain sharp focus.
- They're also smaller and more resistant to shock and vibration than mechanical autofocus lenses of similar specification.



Covers

- A camera cover, also called a housing or enclosure, protects the expensive machine vision camera from the dust, oil, water, food particles, impacts, and other hazards that come with operating in an industrial environment.

Covers

- Covers can also dissipate heat, protect cable connections, and prevent tampering.
- Covers are rated by how well they protect the camera, with IP67 being the usual rating for a dustproof camera in an industrial environment. An IP67 rating also means the enclosure is waterproof, so the operating area can be washed down safely.